MTH202 MIDSEM 2

Exercise 1 (2 points). Answer in short. No explanation required:

(1) What is the order of 3 modulo 10?

(2) The number 2027 happens to be a prime. What is the remainder left by 2025! when divided by 2027?

Exercise 2 (3 points). Compute $3^{987654321} \pmod{164}$.

Exercise 3 (4 points). Find the smallest +ve integer x such that $x \equiv 3^{1003} \pmod{25}$ and $x \equiv (28!)^2 \pmod{29}$.

Exercise 4 (5 points). Show that

(1) The prime factors of $n^2 + n + 1$ are either 3 or are of the form $3k + 1$.

(2) Show the converse: for $p = 3$ or for a prime p of the form $3k + 1$, there is an integer n such that $p \mid n^2 + n + 1$.

Exercise 5 (6 points). Let $\sigma_s(n) := \sum_{d|n} d^s$ be the sum of s-th powers of divisors of n.

(1) Show that σ_s is a multiplicative function by writing it as a convolution of two multiplicative functions.

(2) Prove that if $p_1, \ldots, p_r$ are distinct primes, then:

$$\sigma_s(p_1^{k_1} \cdots p_r^{k_r}) = \left(\frac{p_1^{s(k_1+1)} - 1}{p_1^s - 1} \right) \cdots \left(\frac{p_r^{s(k_r+1)} - 1}{p_r^s - 1} \right)$$

(3) Show that

$$\sum_{d|n} \sigma_1(d) = \sum_{d|n} \frac{n}{d} \sigma_0(d).$$

"I shall be telling this with a sigh
Somewhere ages and ages hence:
Two numbers I had to face,
I took the one in a lower base,
And that has made all the difference."

Make your choices carefully!
